Case 06: Design and Development of a Patient-Specific Infraorbital Rim Reconstruction Implant

1. Introduction

The infraorbital rim is the inferior bony margin of the orbital cavity and plays a critical role in maintaining facial symmetry, orbital support, and eye positioning. Defects in this region may occur due to maxillofacial trauma, tumor resection, congenital deformities, or orbital fractures.

Patient-specific infraorbital rim implants are designed using CT imaging and 3D reconstruction techniques to accurately restore the anatomical contour of the orbital region.

2. Need for Patient-Specific Implant (PSI)

- Restoration of facial symmetry
- Reconstruction of orbital anatomy
- Prevention of enophthalmos (sunken eye appearance)
- Improved aesthetic outcome
- Reduced surgical adjustment
- Better anatomical fit

3. Objective

- To design a patient-specific infraorbital rim reconstruction implant.
- To reconstruct the orbital defect using mirror anatomy.
- To incorporate perforations for weight reduction and tissue integration.

4. Software Used

Component	Software
CT Segmentation	D2P / 3D Slicer
Anatomical Reconstruction	Autodesk Meshmixer
Freeform Implant Design	Meshmixer
Pattern Generation	Meshmixer
STL Generation	Meshmixer
Additive Manufacturing	SLM

5. Methodology

CT Scan Acquisition

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Segmentation of Facial Skeleton

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3D Reconstruction

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Defect Identification

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Mirror Reconstruction of Healthy Side

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Infraorbital Rim Implant Design

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Thickness Generation

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Perforation Pattern Creation

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Boundary Refinement

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STL Export

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SLM Manufacturing

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Post Processing

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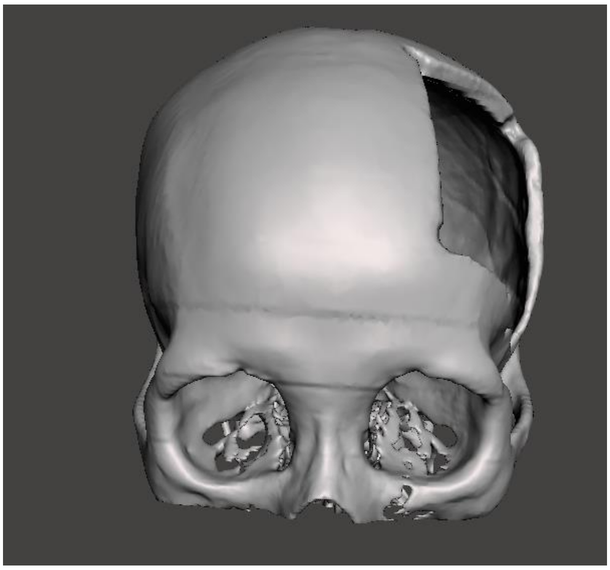
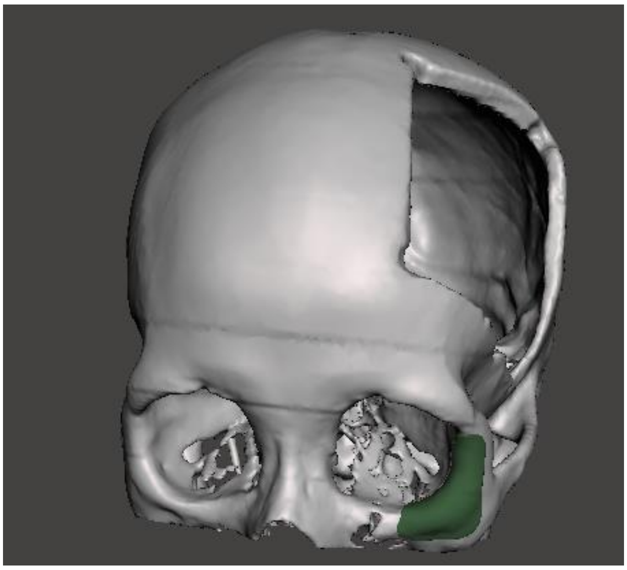
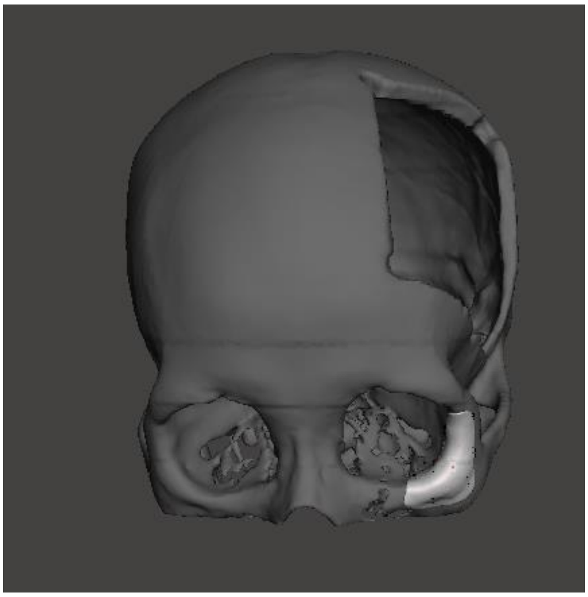
Final PSI Implant







6. Design Features

Feature	Function
Patient-Specific Geometry	Accurate anatomical fit
Mirror Reconstruction	Restoration of missing anatomy
Perforated Structure	Weight reduction
Multiple Fixation Areas	Stable fixation
Thin Profile	Mimics native orbital bone
Anatomical Curvature	Better facial contour restoration

7. Materials Used

Component	Material	Reason for Selection
Infraorbital Rim Implant	Ti-6Al-4V ELI	High strength and biocompatibility
Alternative Material	Medical Grade PEEK	Lightweight and radiolucent
Fixation Screws	Titanium Alloy	Corrosion resistance and stability

8. Design Specifications of PSI Infraorbital Rim Implant

Parameter	Specification
Implant Type	Patient-Specific Infraorbital Rim Implant
Design Method	Mirror Reconstruction
Material	Ti-6Al-4V ELI
Implant Thickness	0.8–1.5 mm
Structure	Perforated Design
Perforation Diameter	1–3 mm
Fixation Method	Titanium Screws
Manufacturing Method	Selective Laser Melting (SLM)
File Format	STL

9. Importance of the Implant

Clinical Importance	Benefit
Orbital Support	Maintains eye position
Facial Symmetry Restoration	Improved aesthetics
Defect Reconstruction	Restores missing bone
Reduced Surgical Time	Better implant fit
Lightweight Design	Improved patient comfort
Patient-Specific Fit	Enhanced clinical outcome

10. Additive Manufacturing Compatibility

Parameter	Specification
Material	Ti-6Al-4V ELI
Technology	Selective Laser Melting (SLM)
Layer Thickness	30–60 µm
Post Processing	Support Removal and Polishing
Sterilization	Medical Grade Sterilization

11. Result

A patient-specific infraorbital rim reconstruction implant was successfully designed using Meshmixer. The implant accurately restored the anatomical contour of the orbital rim and incorporated a perforated structure for weight reduction and improved tissue integration. The final design was exported as an STL file and prepared for additive manufacturing.